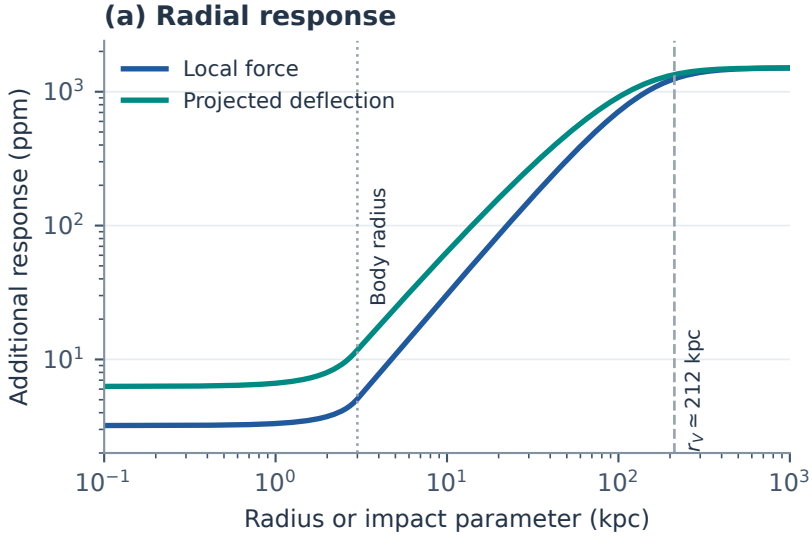

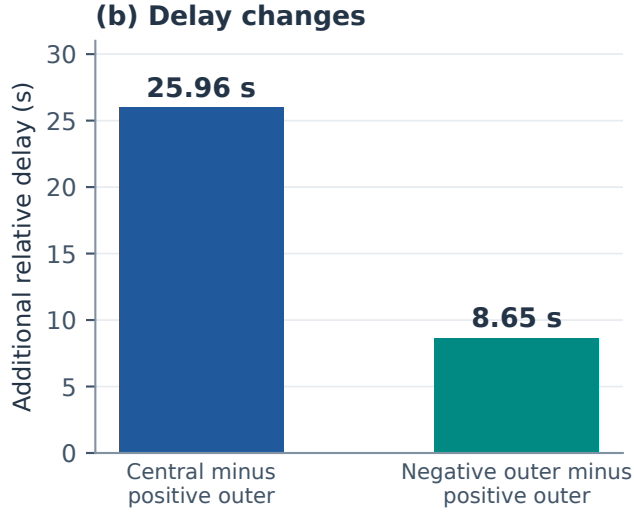

Frozen R1 cubic model: $M = 10^{11} M_\odot$, $R = 3$ kpc, $z_l = 0.3$, $z_s = 1$; matched leading-order comparison.